

## RESEARCH ARTICLE

# Modeling, Simulation, and Experimental Validation: Optimal Placement of a Piezoelectric Energy Harvester via Magnetic Coupling From AC Power Lines Considering Air and Strain Damping

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## ABSTRACT

The growing need for reliable, maintenance-free monitoring of power line infrastructure has intensified interest in sustainable power sources for wireless sensors. Piezoelectric energy harvesting offers a promising alternative to batteries. However, most existing power line harvesters deliberately ignore the combined effects of air damping, strain rate damping, and optimal positioning relative to AC power lines, which limits their power output and real-world applicability. This study addresses the gap and explicitly integrates them within a unified analytical, numerical, and experimental framework validating an optimally positioned piezoelectric energy harvester that extracts mechanical energy from the alternating magnetic field generated by current-carrying conductors. A coupled electromechanical model was formulated, incorporating magnetic excitation, air damping, strain damping to support and verify the results from the experimental setup. Results show that a 6.35 mm<sup>3</sup> magnet placed beneath the power line, generates a peak force of 5.386 mN and an output voltage of 563.68 mV, with the harvested power increasing linearly with airflow velocity under a 9 A current flow at 4 mm from the AC power Line. The LTC3588-1 interface subsequently rectified and boosted the harvested voltage to a stable 3.3 V DC supply, demonstrating suitability for powering low-power Internet of Things (IoT) sensors in smart grid applications.

## 1 | Introduction

The increasing demand for monitoring critical infrastructure, such as power lines, has spurred the development of sustainable power sources for wireless sensors. Traditional monitoring systems rely on batteries, which face several limitations, including limited lifespan, periodic maintenance requirements, and environmental concerns related to disposal. Energy harvesting technologies, particularly piezoelectric methods, offer a promising alternative by utilizing ambient energy sources to

power low-power devices [1]. This approach provides continuous, maintenance-free power for wireless sensors deployed in remote or difficult-to-access areas. Piezoelectric energy harvesting is particularly suitable for environments with prevalent mechanical vibrations, acoustic noise, or magnetic fields. In the case of power lines, the Alternating Current (AC) produces a surrounding magnetic field, which can be used as a source of mechanical energy [2]. Several studies have demonstrated the harvesting techniques using piezoelectric and magnetically coupled mechanism in the case of power lines. A swinging piezoelectric energy harvester